

FACULTY OF ELECTRICAL AND COMPUTER ENGINEERING

Software Engineering I — 2026 Second Term

CogniSync

Software Requirements Specification

Version 1.0

Team 1

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REVISION HISTORY

Table 1 Revision history

Date	Version	Description	Author
2026-10-26	0.1	Document skeleton; functional requirements generated from Azure Boards	Team 1
2026-10-30	0.2	Results of the refinement meeting with the client added to the elicitation log	Team 1
2026-11-03	1.0	Context diagram and feature model; elicitation chapter; 25 non-functional requirements with validation criteria; external interfaces; web and mobile prototype; contributions and authorship declaration	Team 1

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CHAPTER 1

INTRODUCTION

1.1 Purpose

This document specifies the software requirements of CogniSync, a software-as-a-service platform for mental-health clinics. It is the agreement between the development team (Team 1) and the client, Ing. Fabrizzio Ontaneda, about what the system must do (functional requirements) and the qualities and constraints it must satisfy (non-functional requirements). Its intended readers are the client, who validates it and signs the conformity act in Appendix A; the development team, who plan and verify their work against it; and the course instructor.

1.2 Scope

CogniSync centralizes the digital record of each patient, lets therapists capture post-session clinical notes by typed text, voice dictation or a photo of handwritten notes, structures those notes with artificial intelligence into the standard SOAP and DAP clinical formats, recalls the previous session when a new note is started, charts the evolution of each patient's symptoms, and warns about patients at risk of dropping out of treatment. The system has two client components: a **web application** for computers and a **mobile application** for phones and tablets, both served by one back-end API (Section 2.1).

The main benefit expected by the client is that a therapist completes the documentation of a session in no more than three minutes per patient [1]. What the system will not do is listed in Section 2.6.

1.3 Definitions, acronyms and abbreviations

Terms and acronyms are defined in Chapter 7.

1.4 References

The requirements were written following ISO/IEC/IEEE 29148 [2] and IEEE 830 [3]. User stories follow Cohn [4]; non-functional requirements are classified according to Sommerville [5], and their validation criteria follow the fit-criterion idea of the Volere template [6, 7]. The full list is in the References section.

1.5 Overview

Chapter 2 describes the product in context. Chapter 3 explains how the requirements were elicited and managed. Chapter 4 lists the functional requirements, generated from the team's product backlog in Azure Boards so the document and the backlog never disagree. Chapter 5 lists the non-functional requirements with their validation criteria, and Chapter 6 the external interfaces. Appendix A contains the client's signed conformity act and Appendix B the prototype screens and window flow.

CHAPTER 2

OVERALL DESCRIPTION

2.1 Product perspective

CogniSync is a new, self-contained product. It replaces the paper notebooks, spreadsheets and messaging conversations in which clinics keep patient information today.

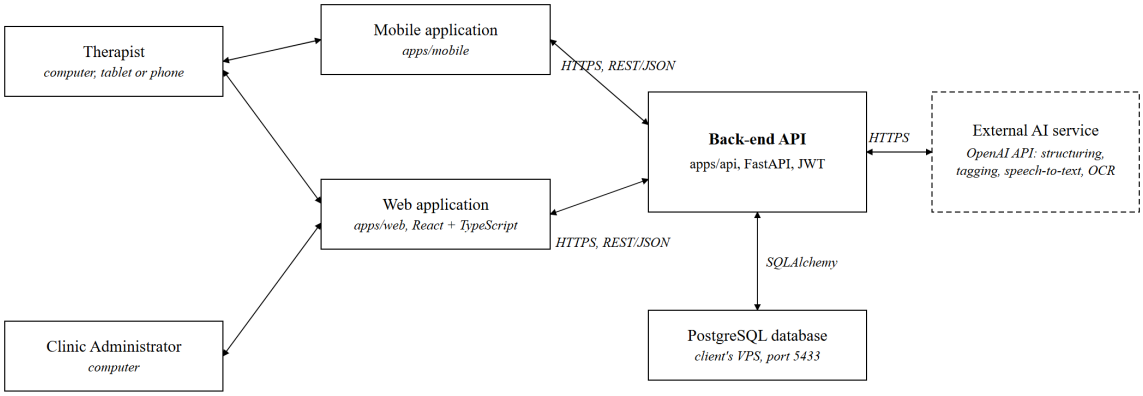
The system has three parts deployed by the team and one external service:

- **Web application** for computers: complete management of patient records, the appointment schedule, note history, evolution charts and clinic administration.
- **Mobile application** for phones and tablets: the therapist's tool after each session — today's agenda, attendance, dropout alerts, and note capture by text, voice or photo.
- **Back-end API** (FastAPI) with a PostgreSQL database, hosted on the client's VPS.
- **External AI service** (OpenAI API) that transcribes, reads and structures notes, called only by the back end.

Figure 2.1 shows the context of the system. Both client applications talk to the API only through HTTPS; neither talks to the database or to the AI service directly.

2.2 Product functions

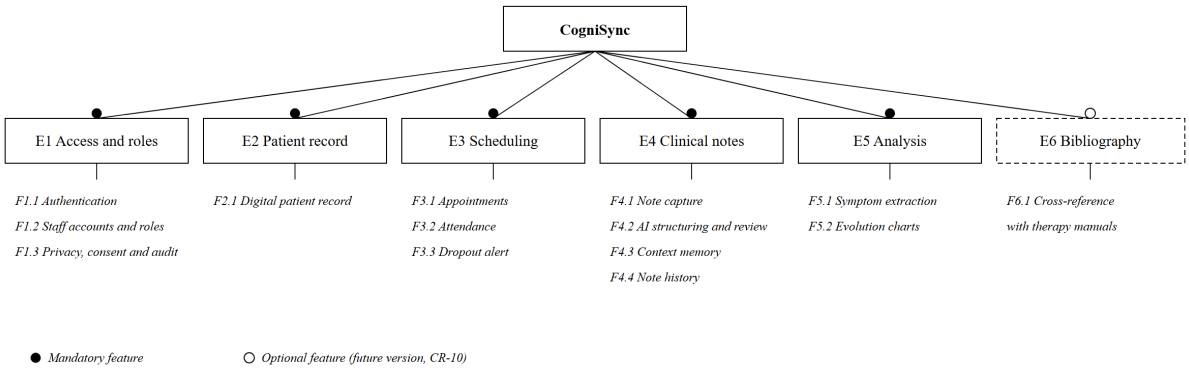
The functions are grouped in the five modules of the concept of operations [1]. Table 2.1 shows which component offers each module.



Solid boxes: users and components built by the team. Dashed box: external service. The AI service receives no patient identifiers (NFR-15).

Figure 2.1 Context diagram of CogniSync
Source: Prepared by the team

Figure 2.2 shows the feature model: the six epics of the product backlog and the features inside each one. Every epic is part of this release except the bibliographic cross-reference, which the client marked as future (`CR-10`).



Each epic lists its features as they are recorded in Azure Boards. The functional requirements in Chapter 4 follow this structure.

Figure 2.2 Feature model of CogniSync
Source: Adapted from CogniSync Team 1 [1, Figure 1] and the epics and features in Azure Boards

Table 2.1 Functions per module and component

Module	Web application	Mobile application
Patients and scheduling	Full patient-record management, appointment scheduling, attendance, dropout-alert panel	Record lookup, today's agenda, attendance, dropout alerts
Clinical notes	Typed notes, review of the structured note, context memory, note history	Typed, dictated and photographed notes, review of the structured note, context memory
Analysis and evolution	Heat maps and trend lines; bibliographic cross-reference (future version)	—
Clinic administration	Staff accounts and roles	—
Cross-cutting	Sign-in, role-based access, secure connection	Sign-in, role-based access, secure connection

Source: Adapted from CogniSync Team 1 [1, §4.2]

2.3 User characteristics

Clinic Administrator. Manages the clinic's staff accounts and roles and consults the lists of staff and patients. Has no access to clinical content. Intermediate technical skill; uses the system periodically.

Therapist. Psychologist or mental-health professional; the primary user. Manages their patients' records, records attendance, writes and reviews clinical notes, and follows each patient's evolution. Basic technical skill, so the interface must be minimal and fast; uses the system after every session.

2.4 Constraints

The client fixed the technology of the solution (`CR-14` to `CR-18` in the client's requirements document): a monorepo separating the clients from the API; React with TypeScript and Tailwind CSS for the web client; FastAPI (Python) for the back end, with JWT authentication and calls to the OpenAI API; PostgreSQL accessed through SQLAlchemy; and deployment on an Ubuntu VPS behind Nginx. Chapter 5 states these as organizational (development) requirements.

The client did not fix the technology of the mobile application. The team proposes React Native with Expo and TypeScript in `apps/mobile`, sharing the API types and the `httpClient.ts` wrapper with the web application; the proposal is pending the client's confirmation (`OQ-24`). The product is also subject to Ecuador's Organic Law on Personal Data Protection, which treats health data as a special category [8] (NFR-23).

2.5 Assumptions and dependencies

Assumptions.

- Therapists have an internet connection while they capture and structure a note; the system does not work offline.
- To dictate or photograph a note, the therapist's phone or tablet has a working microphone and camera, and the user grants the application permission to use them.
- The clinic provides the VPS where the system runs and pays for the external AI service.
- Every patient whose notes are processed by the AI service has given informed consent, recorded in the patient record (US-28).

Dependencies.

- The OpenAI API, its availability and its terms of service. If the service fails, typed notes are still saved (NFR-11).
- The speech-to-text and handwriting-recognition services chosen in SPIKE-01 and SPIKE-02. US-13 and US-14 depend on their results.
- The client's answers to the open questions (`OQ-xx`). Proposed values marked in this document are confirmed or corrected when the client signs the conformity act (Appendix A).

2.6 Out of scope

The system will not: issue diagnoses or treatment recommendations automatically; suggest intervention plans from therapy manuals (planned for a future version); handle billing or payments; offer video calls or online consultations; offer a portal for patients; or integrate with insurers or other external health systems [1, §2.2.2].

CHAPTER 3

REQUIREMENTS ELICITATION AND METHODOLOGY

The team combined several elicitation techniques, as Sommerville [5, §4.3] recommends, because no single technique uncovers every requirement. Table 3.1 summarizes them. The evidence of each one (documents, minutes, drafts and the prototype) is in the `docs/elicitation` folder of the public repository [9].

Table 3.1 Elicitation techniques used

Technique	Date	Participants	Result
Document analysis	2026-10-07	Team	18 client requirements with stable IDs (`CR-01` to `CR-18`) and the first list of gaps
Concept of operations and feature modeling	2026-10-16	Team	CONOPS v1.0: stakeholders, 5 modules and the feature model
User-story workshop	2026-10-26	Team	First backlog: 23 stories in 6 epics, prioritized with MoSCoW
Instructor review	2026-10-26	Instructor, teaching assistant, team	Feedback: more detail and measurable acceptance criteria
Backlog refinement	2026-10-26	Team	Backlog v2: 30 stories, 14 features, Given/When/Then criteria, 24 open questions
Refinement meeting with the client	2026-10-30	Client, team	Stories reviewed one by one; decisions listed in Section 3.5
Prototyping	2026-11-03	Team	10 web screens and 4 mobile screens (Appendix B)

Source: Elicitation log in the public repository [9]

3.1 Document analysis

The starting point was the requirements document written and electronically signed by the client. The team transcribed it into English, gave each requirement a stable identifier (`CR-xx`) and marked every term that had no measure or definition. The analysis found three kinds of problems: *ambiguities* (“high risk of dropout”, “brief summary”, “minimalist and intuitive”), *omissions* (appointments and note history, although `CR-04` and `CR-07` depend on them; account recovery; consent to process health data) and *requirements that are not measurable* (“the web system must be hosted on a VPS” states a platform, not an availability level). Each problem became a gap story or an open question.

3.2 Concept of operations and feature modeling

The concept of operations [1] described the stakeholders, the current way clinics work, the proposed system and its five modules. Its feature model (Figure 2.2) fixed which functions belong to the web application and which to the mobile application, and became the epic and feature structure of the product backlog.

3.3 User-story workshop

The team wrote the first version of the backlog as user stories in the form proposed by Cohn [4]: 23 stories grouped into six epics, each with acceptance criteria, a MoSCoW priority, its source in the client’s document and a justification of how it was sliced. Stories were checked against the INVEST criteria and ordered by the path a therapist follows in a working day (story mapping). The workshop also produced a questionnaire of 31 questions for the client, the origin of the open questions in Section 3.7.

3.4 Instructor review and backlog refinement

On 2026-10-26 the instructor and the teaching assistant reviewed the progress of the user stories. Their feedback was that criteria must be more detailed, measurable and specific. The team refined the backlog the same day: stories were translated into English, criteria were rewritten as Given/When/Then scenarios with numbers and exact messages, 14 features were added from the feature model, seven gap stories were created (US-24 to US-30), dependencies were linked, and every value the team proposed was recorded as an open question instead of being treated as confirmed.

3.5 Refinement meeting with the client

On 2026-10-30 the team met the client in a recorded Microsoft Teams meeting and went through the backlog story by story. The meeting produced these results:

- The summary of the previous session keeps a limit of 5 lines, is built only from approved notes, and says when there is no previous session (US-18).
- The structured note is always shown next to the therapist's original text, which is never lost (US-15, US-17).
- Notes are listed from the most recent to the oldest with their date, format and status; opening one shows the original text and the structured version; notes cannot be deleted (US-19).
- The therapist can add, edit or remove symptom tags, and the corrections are reflected in the patient's evolution chart, which uses approved notes only (US-21, US-22).
- Every change to a patient record keeps who made it and when (US-07).
- Signing in with a Google or Microsoft account is left for a future version.
- The team will present the prototype at the next meeting, where the client signs the conformity act.

3.6 Prototyping

The team built a navigable prototype of 10 web screens and 4 mobile screens with synthetic data (Appendix B). Its purpose is to validate with the client the content of each screen and the flow between them before any code is written. Its layout was informed by real products of the same category, collected from the Mobbin design library [10]: clinical note takers for therapists, sign-in pages with third-party accounts, team-management settings and trend charts.

3.7 Open questions

When a criterion needed a value that the client never gave (a time limit, a threshold, a list of fields), the team proposed one so the criterion could be tested, wrote the question in the open-questions list of the repository, and marked the requirement with its identifier (`OQ-nn`). Twenty-nine questions were recorded, OQ-25 to OQ-29 while writing the non-functional requirements of Chapter efch:nfr. One of them (`OQ-14`, web and mobile application) is closed. The rest are confirmed or corrected by the client when the conformity act is signed.

3.8 Methodology

The team works with Scrum in two-week sprints. Requirements are managed as a product backlog in Azure Boards with four levels: epics (the modules of the concept of operations), features, product backlog items written as user stories, and tasks.

Each user story follows the Connextra form (*As a …, I want …, so that …*) [4] and has acceptance criteria written as Given/When/Then scenarios. A story is ready for a sprint only when every criterion is measurable: it states a number, an explicit rule or an observable outcome, and contains none of the vague words listed in the team's working agreement. Values the team proposed but the client has not yet confirmed are recorded as open questions and marked in the requirement.

Prioritization. Every story has a MoSCoW priority: *Must* (delivered in this release), *Should* (expected), *Could* (desirable if time allows) and *Won't* (not in this release). The backlog holds 13 *Must*, 13 *Should*, 3 *Could* and 1 *Won't* story.

Definition of Done. A story is done when every acceptance criterion passes and is shown in the sprint review, the code is merged into the main branch through a reviewed pull request linked to the work item, automated tests cover the criteria where feasible, and the cross-cutting requirements hold: UUID identifiers, no real patient data, screens usable at 360 px and 1280 px, and permissions enforced by the API. Each sprint ends with a review with the client, its minutes and an acceptance form signed by the client.

Traceability. Each client requirement (`CR-xx`) is linked to the stories that implement it (Table 4.1); each story has a stable key (`US-nn`) and an Azure Boards number (`AB#n`); and each commit and pull request cites that number, so Azure DevOps links the code to the requirement. Chapter 4 is generated from Azure Boards, so the document and the backlog never disagree.

CHAPTER 4

FUNCTIONAL REQUIREMENTS

This chapter is generated from the product backlog in Azure Boards by the script `tools/srs/export_fr.py`. To change a requirement, edit its work item, not this text. Requirements are grouped by epic and feature. Each requirement shows:

ID the stable key of the user story (`US-nn`) and its Azure Boards number (`AB#n`);

Statement the user story;

Priority MoSCoW: *Must*, *Should*, *Could* or *Won't* (this release);

Platform the component that implements it: web application, mobile application or both;

Source where the requirement comes from: a client requirement (`CR-xx`), the concept of operations, a gap found by the team, or a team proposal;

Depends on stories that must be delivered first;

Acceptance criteria the conditions the client will use to accept the story;

Pending open questions (`OQ-nn`) whose proposed values the client still has to confirm.

4.1 Access and roles

4.1.1 Authentication

Secure sign-in, sign-out, session expiry and password recovery. CONOPS §4.7.1, CR-02.

US-01 Sign in	AB#33
Statement	As a Therapist, I want to sign in with my email and password, so that only I can reach my patients' clinical information.
Priority	Must
Platform	Web and Mobile
Source	Client requirement; CR-02 (secure authentication)
Depends on	—
Acceptance criteria	AC1. **Given** an active account and correct credentials, **when** the user signs in, **then** a Therapist lands on their patient dashboard and a Clinic Administrator lands on Staff management. AC2. **Given** a wrong email or a wrong password, **when** the user signs in, **then** the message "Invalid credentials" is shown without saying which field was wrong. AC3. **Given** a deactivated account (US-04), **when** it signs in with correct credentials, **then** access is denied with "Account deactivated. Contact your clinic administrator." AC4. **Given** 5 consecutive failed attempts on one account (OQ-01), **when** another attempt is made within the next 15 minutes, **then** it is rejected even with correct credentials and the time of unlock is shown. AC5. **Given** any request to http://, **when** it reaches the server, **then** it is redirected to https:// (CR-11).
Pending	Client confirmation of OQ-01

US-02 Sign out and session expiry	AB#34
Statement	As a Therapist, I want to sign out and have my session expire after a period without use, so that nobody can see patient records if I leave the device unattended in my office.
Priority	Must
Platform	Web and Mobile
Source	Gap found by the team; required by a secure sign-in for mental-health data (derived from CR-02)
Depends on	US-01

US-02 (continued)

Acceptance criteria	AC1. **Given** a signed-in user, **when** they select “Sign out”, **then** the sign-in page is shown and the browser Back button shows no patient data.
	AC2. **Given** 30 minutes without activity (OQ-02), **when** the user performs any action, **then** they are asked to sign in again.
	AC3. **Given** a note being edited when the session expires, **when** the same user signs in again, **then** the note is restored as a draft containing the text from the last autosave (at most 30 s old, US-12).
	AC4. **Given** a session that was signed out, **when** its old token is sent to the API, **then** the API answers 401 Unauthorized.
Pending	Client confirmation of OQ-02

US-24 Reset a forgotten password	AB#72
Statement	As a Therapist, I want to reset my password through my email, so that I can regain access without waiting for the administrator.
Priority	Should
Platform	Web and Mobile
Source	Team proposal; team proposal (complements CR-02)
Depends on	US-01
Acceptance criteria	AC1. **Given** the sign-in page, **when** the user selects “Forgot password?” and submits an email, **then** the same message “If the email is registered, you will receive a link” is shown whether or not the email exists.
	AC2. **Given** a registered email, **when** the request is made, **then** a single-use reset link valid for 30 minutes is sent to it (OQ-23).
	AC3. **Given** a link that was already used or is older than 30 minutes, **when** it is opened, **then** “This link has expired” is shown and the password is not changed.
	AC4. **Given** a new password shorter than 10 characters (OQ-23), **when** the user submits it, **then** it is rejected with the minimum length stated.
Pending	Client confirmation of OQ-23

4.1.2 Staff accounts and roles

The Clinic Administrator manages therapist accounts and roles. CONOPS §3.1.1, §4.6, CR-01.

US-03 Register therapists	AB#35
Statement	As a Clinic Administrator, I want to register the therapists of my clinic, so that they can use the system with the correct role.
Priority	Must
Platform	Web
Source	Client requirement; CR-01 (role management); CONOPS §3.1.1(a)
Depends on	US-01
Acceptance criteria	AC1. **Given** the Staff management page, **when** the administrator submits first names, last names, email and role, **then** the account is created and can sign in. AC2. **Given** an email that is already registered, **when** the administrator submits it, **then** "This email is already registered" is shown and no account is created. AC3. **Given** a signed-in Therapist, **when** they look for Staff management in the menu or type its URL, **then** the menu entry is absent and the URL shows "Access denied" (API answers 403).

US-04 Deactivate and reactivate therapists	AB#36
Statement	As a Clinic Administrator, I want to deactivate and reactivate a therapist's account, so that someone who leaves the clinic can no longer see patient records.
Priority	Should
Platform	Web
Source	Client requirement; CR-01 (role management); CONOPS §3.1.1(a)
Depends on	US-03
Acceptance criteria	AC1. **Given** an active therapist, **when** the administrator deactivates the account, **then** the therapist cannot sign in and any open session is rejected on its next request. AC2. **Given** a deactivated therapist, **when** the administrator checks the data, **then** 100 % of that therapist's patient records and notes are still stored and unchanged. AC3. **Given** a deactivated therapist, **when** the administrator reactivates the account, **then** the therapist can sign in and sees the same patients as before deactivation.

US-25 Edit a staff account and its role — AB#73

Statement	As a Clinic Administrator, I want to edit a staff account's name, email and role, so that access stays correct when someone's details or responsibilities change.
Priority	Should
Platform	Web
Source	Client requirement; CR-01; CONOPS §3.1.1(a, b), §4.6.2
Depends on	US-03
Acceptance criteria	AC1. **Given** a staff account, **when** the administrator changes its names or email and saves, **then** the new values appear in the staff list. AC2. **Given** an email already used by another account, **when** the administrator saves it, **then** "This email is already registered" is shown and nothing changes. AC3. **Given** a role changed from Therapist to Clinic Administrator, **when** that user next signs in, **then** they see Staff management and no clinical screens (OQ-18). AC4. **Given** the only active Clinic Administrator, **when** someone tries to change that account's role or deactivate it, **then** the action is blocked with "The clinic needs at least one active administrator".
Pending	Client confirmation of OQ-18

US-26 See the clinic's staff and patient lists — AB#74

Statement	As a Clinic Administrator, I want to see the list of therapists and how many patients each one has, so that I can supervise the clinic without accessing clinical content.
Priority	Should
Platform	Web
Source	Gap found by the team; CONOPS §3.1.1(c) (supervise operation), not in v1
Depends on	US-03
Acceptance criteria	AC1. **Given** the Staff page, **when** it loads, **then** each therapist is listed with name, email, status (Active or Deactivated) and number of active patients. AC2. **Given** the clinic patient list, **when** the administrator opens it, **then** each row shows only the patient's name, assigned therapist and status (Active or Archived) (OQ-18). AC3. **Given** a Clinic Administrator, **when** they request a patient record or note through the UI or the API, **then** access is denied (API answers 403) (OQ-18).
Pending	Client confirmation of OQ-18

4.1.3 Privacy, consent and audit

Who accessed what, patient consent for AI processing, non-enumerable IDs, HTTPS. CONOPS §4.7.3, CR-11, executive summary (problem 1).

US-27 Audit access to patient records	AB#75
Statement	As a Clinic Administrator, I want a log of who viewed or changed each patient record and note, so that the clinic can prove that only authorized staff accessed sensitive data.
Priority	Must
Platform	Web
Source	Gap found by the team; executive summary, problem 1 (“no record of who consults each file”); CR-11; CONOPS §4.7.3
Depends on	US-01
Acceptance criteria	AC1. **Given** any view, creation, edit or archive of a patient record or note, **when** it happens, **then** a log entry stores the UTC timestamp, user, action and record identifier. AC2. **Given** the audit page, **when** the administrator filters by patient, user or date range, **then** only matching entries are listed, newest first. AC3. **Given** any role, **when** it tries to edit or delete a log entry through the UI or the API, **then** the action is impossible (no UI option; API answers 403). AC4. **Given** a Therapist, **when** they request the audit page, **then** access is denied (OQ-16). AC5. **Given** a log entry, **when** 5 years have passed since it was written, **then** it is still retrievable (OQ-16).
Pending	Client confirmation of OQ-16

US-28 Record the patient’s consent for AI processing	AB#76
Statement	As a Therapist, I want to record whether the patient consented to AI processing of their notes, so that the clinic complies with Ecuador’s personal data protection law (LOPDP) for health data.
Priority	Should
Platform	Web
Source	Team proposal; LOPDP (health data is a special category); executive summary (main risk: sensitive data with AI)
Depends on	US-05

US-28 (continued)

Acceptance criteria	AC1. **Given** a patient record, **when** it is created, **then** the field "AI processing consent" exists with the value "Not granted" and the therapist can set "Granted" with its date (OQ-15). AC2. **Given** a patient without consent, **when** the therapist opens a note, **then** voice dictation, photo capture, SOAP/DAP structuring and symptom tagging are disabled with the reason shown, and typed notes still work. AC3. **Given** consent that is withdrawn, **when** the therapist saves the change, **then** no new AI processing occurs for that patient from that moment and the change is written to the audit log (US-27).
Pending	Client confirmation of OQ-15

4.2 Patient record

4.2.1 Digital patient record

Create, list, search, edit and archive patient records. CONOPS §4.3.1, CR-03.

US-05 Create a patient record AB#37

Statement	As a Therapist, I want to create a record for a new patient with their demographics and consultation history, so that all their information is in one place from the first session.
Priority	Must
Platform	Web
Source	Client requirement; CR-03 (digital record: create)
Depends on	US-01
Acceptance criteria	AC1. **Given** the New patient form, **when** the therapist saves it with first names, last names, date of birth, sex, phone and reason for consultation filled in (OQ-03), **then** the record is created with a UUID identifier (CR-11). AC2. **Given** a missing required field, **when** the therapist saves, **then** that field is marked "Required" and nothing is saved. AC3. **Given** a date of birth later than today, **when** the therapist saves, **then** "Date of birth cannot be in the future" is shown and nothing is saved. AC4. **Given** a record that was just created, **when** the therapist opens their patient list, **then** the new patient appears in it.

Pending	Client confirmation of OQ-03

US-06 List and search patients — AB#38

Statement	As a Therapist, I want to see my patient list and search it by name, so that I can open the right record without scrolling through the whole list.
Priority	Must
Platform	Web and Mobile
Source	Client requirement; CR-03 (digital record: read)
Depends on	US-05
Acceptance criteria	AC1. **Given** the dashboard, **when** it loads, **then** it lists the therapist's active patients sorted by last name from A to Z. AC2. **Given** the search box, **when** the therapist types "per", **then** only patients whose first or last name contains "per" are listed, ignoring case and accents ("Pérez" matches). AC3. **Given** a patient of another therapist, **when** the list or search results load, **then** that patient never appears (OQ-04). AC4. **Given** a therapist with 500 patients, **when** they type a search term, **then** results appear in 1 second or less (OQ-22).
Pending	Client confirmation of OQ-04, OQ-22

US-07 Edit a patient record — AB#39

Statement	As a Therapist, I want to update a patient's data, so that the record reflects their current situation.
Priority	Must
Platform	Web
Source	Client requirement; CR-03 (digital record: update)
Depends on	US-05
Acceptance criteria	AC1. **Given** a saved change, **when** the therapist reopens the record, **then** the new value is shown. AC2. **Given** a date of birth later than today, **when** the therapist saves, **then** the change is rejected with "Date of birth cannot be in the future". AC3. **Given** any saved change, **when** the record is opened, **then** it shows "Last modified <date and time> by <user>".

US-08 Archive a patient		AB#40
Statement	As a Therapist, I want to archive a patient who finished or abandoned treatment, so that my list stays organized without losing their clinical history.	
Priority	Must	
Platform	Web	
Source	Client requirement; CR-03 (digital record: archive)	
Depends on	US-05	
Acceptance criteria	AC1. **Given** an archived patient, **when** the main patient list loads, **then** the patient is not in it.	
	AC2. **Given** an archived patient, **when** the therapist opens their notes, **then** every note is still stored and readable.	
	AC3. **Given** the "Archived" filter, **when** the therapist applies it, **then** the patient is listed and can be reactivated, after which they appear in the main list again.	
	AC4. **Given** any role, **when** looking for a way to delete a patient, **then** no delete option exists in the UI and the API rejects DELETE requests on patients.	

4.3 Scheduling and attendance

4.3.1 Appointments

Schedule appointments and see the day's agenda. CONOPS §4.3.2.

US-09 Schedule appointments		AB#41
Statement	As a Therapist, I want to schedule each patient's appointments with date and time, so that their attendance can be tracked.	
Priority	Should	
Platform	Web	
Source	Gap found by the team; the client module is called "Patients and Scheduling" but has no scheduling requirement; CR-04 needs appointment history. CONOPS §4.3.2	
Depends on	US-05	

US-09 (continued)

Acceptance criteria	AC1. **Given** a patient record, **when** the therapist creates an appointment with date, start time and duration, **then** it is saved with status "Scheduled" and the duration defaults to 50 minutes (OQ-05). AC2. **Given** an appointment that overlaps another one of the same therapist, **when** the therapist saves it, **then** a warning shows the other appointment's patient and time and saving requires confirmation. AC3. **Given** a saved appointment, **when** the patient record is opened, **then** the appointment is listed with status "Scheduled".
Pending	Client confirmation of OQ-05

US-29 See today's agenda on my phone	AB#77
Statement	As a Therapist, I want to see today's appointments on my phone, so that I can prepare each session and write its note right after it.
Priority	Should
Platform	Mobile
Source	Gap found by the team; CONOPS §4.3.2 and §4.2 (mobile component: today's agenda), not in v1
Depends on	US-09
Acceptance criteria	AC1. **Given** the mobile home screen, **when** it opens, **then** today's appointments are listed by start time with patient name, time and status. AC2. **Given** an appointment in the list, **when** the therapist taps it, **then** the patient's record summary and a "New note" button are shown. AC3. **Given** no appointments today, **when** the screen opens, **then** "No appointments today" is shown. AC4. **Given** a 360 px wide screen, **when** the agenda is shown, **then** it needs no horizontal scrolling (CR-12).

4.3.2 Attendance

Mark appointments as attended, cancelled or no-show. CONOPS §4.3.3.

US-10 Record attendance	AB#42
Statement	As a Therapist, I want to mark each appointment as attended, cancelled or no-show, so that the system has the history it needs to detect dropout risk.
Priority	Should

US-10 (continued)

Platform	Web and Mobile
Source	Gap found by the team; data required by CR-04 (dropout alert). CONOPS §4.3.3
Depends on	US-09
Acceptance criteria	AC1. **Given** an appointment whose start time has passed, **when** the therapist marks it, **then** the options are “Attended”, “Cancelled” and “No-show”. AC2. **Given** an appointment in the future, **when** the therapist marks it, **then** only “Cancelled” is available. AC3. **Given** a patient record, **when** it is opened, **then** it shows the count of appointments per status, for example “Attended 4 · Cancelled 1 · No-show 1”.

4.3.3 Dropout alert

Detect patients at high risk of abandoning therapy. CONOPS §4.3.4, CR-04.

US-11 See a dropout-risk alert	AB#43
Statement	As a Therapist, I want to see an alert on my dashboard when a patient is at high risk of abandoning therapy, so that I can contact them before they leave treatment.
Priority	Should
Platform	Web and Mobile
Source	Client requirement; CR-04 (dropout alert)
Depends on	US-10

US-11 (continued)

Acceptance criteria	AC1. **Given** a patient whose last 2 appointments are “Cancelled” or “No-show” (OQ-06), **when** the therapist opens the dashboard, **then** the patient appears in the “Dropout risk” panel with the reason “2 consecutive missed appointments”. AC2. **Given** a patient who missed 40 % or more of their appointments in the last 60 days with at least 3 appointments in that window (OQ-06), **when** the dashboard loads, **then** the patient appears in the panel with the reason “<n> % missed in 60 days”. AC3. **Given** 5 appointments in 60 days with 2 non-consecutive misses (40 %), **when** the dashboard loads, **then** the alert is shown; given 5 appointments with 1 miss (20 %), then it is not shown. AC4. **Given** a patient who meets neither rule, including after attending their latest appointment, **when** the dashboard loads, **then** the patient is not in the panel.
Pending	Client confirmation of OQ-06

4.4 Clinical notes

4.4.1 Note capture

Typed text, voice dictation, photo of handwritten notes. CONOPS §4.4.1, CR-05.

US-12 Write a note as text AB#44

Statement	As a Therapist, I want to write the session summary as free text, so that I can record what happened without another tool.
Priority	Must
Platform	Web and Mobile
Source	Client requirement; CR-05(a) (manual text input)
Depends on	US-05
Acceptance criteria	AC1. **Given** a patient and a session date, **when** the therapist saves a typed note, **then** the note is stored linked to that patient and date. AC2. **Given** the note editor, **when** the text reaches 5,000 characters (OQ-07), **then** further typing is blocked and the counter shows “5000/5000”. AC3. **Given** a note being typed, **when** 30 seconds pass since the last autosave, **then** a draft is saved; after closing and reopening the browser, the last draft is restored.

US-12 (continued)

Pending	Client confirmation of OQ-07

US-13 Dictate a note by voice — AB#45

Statement	As a Therapist, I want to dictate the note and have it transcribed to text, so that I can record the session without typing.
Priority	Should
Platform	Mobile
Source	Client requirement; CR-05(b) (voice dictation)
Depends on	SPIKE-01, US-12, US-28
Acceptance criteria	AC1. **Given** a browser with microphone permission, **when** the therapist records up to 10 minutes of audio (OQ-08), **then** they receive the transcript as editable text. AC2. **Given** a 5-minute recording, **when** it is submitted, **then** the transcript appears within 60 seconds (OQ-08). AC3. **Given** a recording without recognizable speech, **when** it is submitted, **then** "No speech detected" is shown and any text already in the note is unchanged. AC4. **Given** a transcript that was produced, **when** storage is inspected, **then** the audio file for that note no longer exists (OQ-08).
Pending	Client confirmation of OQ-08

US-14 Create a note from a photo of handwritten notes — AB#46

Statement	As a Therapist, I want to upload a photo of my paper notes and get the text, so that I do not have to retype what I wrote in my notebook.
Priority	Could
Platform	Mobile
Source	Client requirement; CR-05(c) (image upload with OCR)
Depends on	SPIKE-02, US-12, US-28

US-14 (continued)

Acceptance criteria	AC1. **Given** the photo upload, **when** the therapist selects JPG or PNG files of up to 10 MB each, **then** they are accepted; any other format or a larger file is rejected with a message stating the allowed formats and size.
	AC2. **Given** one note, **when** the therapist adds photos, **then** up to 5 are accepted and a 6th is rejected (OQ-09).
	AC3. **Given** the extracted text, **when** it is shown, **then** it is editable before the note is saved.
	AC4. **Given** a photo with no legible text, **when** it is processed, **then** "No readable text found" is shown.
Pending	Client confirmation of OQ-09

4.4.2 AI structuring and review

Structure raw notes into SOAP or DAP; therapist reviews and approves. CONOPS §4.4.2, CR-06.

US-15 Structure a note as SOAP	AB#47
Statement	As a Therapist, I want the system to turn my raw notes into a SOAP note (Subjective, Objective, Assessment, Plan), so that I get a standardized clinical note without writing it myself.
Priority	Must
Platform	Web and Mobile
Source	Client requirement; CR-06 (automatic structuring)
Depends on	US-12, US-28
Acceptance criteria	AC1. **Given** raw note text, **when** the therapist requests SOAP, **then** the result has exactly four sections: Subjective, Objective, Assessment and Plan.
	AC2. **Given** raw text with no information for a section, **when** it is structured, **then** that section reads "No information in the notes".
	AC3. **Given** 10 synthetic test notes, **when** the client reviews the structured results, **then** the client approves at least 8 of them.
	AC4. **Given** raw text of up to 5,000 characters, **when** it is submitted, **then** the structured note appears within 20 seconds (OQ-21).
	AC5. **Given** the AI service fails or does not answer within 30 seconds (OQ-21), **when** the therapist is waiting, **then** "Structuring failed. Your original text is saved." is shown and the original text is unchanged.
	AC6. **Given** any request to the AI service, **when** its payload is inspected, **then** it contains no patient name or identifier (OQ-17).

US-15 (continued)

Pending	Client confirmation of OQ-17, OQ-21

US-16 Structure a note as DAP — AB#48

Statement	As a Therapist, I want the system to turn my raw notes into a DAP note (Data, Assessment, Plan), so that I can use the format I or my clinic prefer.
Priority	Should
Platform	Web and Mobile
Source	Client requirement; CR-06 (automatic structuring)
Depends on	US-15
Acceptance criteria	AC1. **Given** raw note text, **when** the therapist requests DAP, **then** the result has exactly three sections: Data, Assessment and Plan. AC2. **Given** the structuring screen, **when** the therapist chooses SOAP or DAP and marks one as default, **then** that format is preselected the next time. AC3. **Given** the AI service fails or does not answer within 30 seconds (OQ-21), **when** the therapist is waiting, **then** the same message as in US-15 is shown and the original text is unchanged.
Pending	Client confirmation of OQ-21

US-17 Review and approve a note — AB#49

Statement	As a Therapist, I want to review and edit the structured note before approving it, so that it has no errors, because I am responsible for its content.
Priority	Must
Platform	Web and Mobile
Source	Team proposal; team proposal (AI output must be validated by the professional); measures the 3-minute target of CR-12
Depends on	US-15
Acceptance criteria	AC1. **Given** a structured note, **when** the therapist reviews it, **then** every section is editable. AC2. **Given** a note that is not approved, **when** it is listed, **then** its status is "Draft"; once approved, its status is "Approved" with the date, time and approving user. AC3. **Given** an approved note, **when** the therapist opens it, **then** the text cannot be edited and only a dated addendum can be added (OQ-10). AC4. **Given** 3 therapists writing 5 notes each with synthetic cases, **when** the time from opening a new note to approving it is measured, **then** all 15 cases take 3 minutes or less (CR-12).

US-17 (continued)

Pending	Client confirmation of OQ-10

4.4.3 Context memory

Summary of the previous session when a new note starts. CONOPS §4.4.3, CR-07.

US-18 See the previous session's summary	AB#50
Statement	As a Therapist, I want to see a summary of the previous session when I start a new note for the same patient, so that I can pick up where we left off without rereading the whole record.
Priority	Must
Platform	Web and Mobile
Source	Client requirement; CR-07 (context memory)
Depends on	US-17
Acceptance criteria	AC1. **Given** a patient with approved notes, **when** the therapist starts a new note, **then** a summary of the latest approved note is shown in 5 lines or fewer (OQ-11). AC2. **Given** the summary, **when** it is shown, **then** it states the date of that session. AC3. **Given** a patient with no approved notes, **when** the therapist starts a new note, **then** "First recorded session" is shown.
Pending	Client confirmation of OQ-11

4.4.4 Note history

Chronological list of a patient's notes. CONOPS §4.4.4.

US-19 Browse a patient's note history	AB#51
Statement	As a Therapist, I want to see all of a patient's notes ordered by date, so that I can review their whole process.
Priority	Must
Platform	Web

US-19 (continued)

Source	Gap found by the team; analysis is based on notes, but the client document never says earlier notes can be consulted. CONOPS §4.4.4
Depends on	US-12
Acceptance criteria	AC1. **Given** a patient with notes, **when** the therapist opens the history, **then** notes are listed from newest to oldest with date, format (Text, SOAP or DAP) and status. AC2. **Given** the history, **when** the therapist selects a note, **then** its full content is shown. AC3. **Given** a note of another therapist's patient, **when** a therapist requests it through the UI or the API, **then** access is denied (API answers 403) (OQ-04).
Pending	Client confirmation of OQ-04

4.5 Analysis and evolution

4.5.1 Symptom extraction

AI tags symptoms and their intensity in approved notes; therapist can correct them. CONOPS §4.5.1, CR-08.

US-20 Tag symptoms automatically	AB#52
Statement	As a Therapist, I want the system to detect and tag the symptoms mentioned in each approved note, so that I do not have to classify them myself.
Priority	Should
Platform	Web
Source	Client requirement; CR-08 (symptom extraction)
Depends on	US-17, US-28

US-20 (continued)

Acceptance criteria	AC1. **Given** an approved note that mentions “insomnio” and “ataques de pánico”, **when** it is processed, **then** it has exactly those 2 symptom tags. AC2. **Given** any tag, **when** it is stored, **then** its symptom belongs to the catalog agreed with the client and its intensity is an integer from 1 to 10 (OQ-12). AC3. **Given** an approved note that mentions no symptoms, **when** it is processed, **then** it has 0 tags. AC4. **Given** 10 synthetic test notes, **when** the client reviews the tags, **then** the client approves the tags of at least 8 notes.
Pending	Client confirmation of OQ-12

US-21 Correct symptom tags	AB#53
Statement	As a Therapist, I want to add, remove or correct a note’s symptom tags, so that the evolution charts are based on correct data.
Priority	Could
Platform	Web
Source	Team proposal; team proposal (same reasoning as US-17: the professional validates AI output)
Depends on	US-20
Acceptance criteria	AC1. **Given** a tag the therapist removes, **when** the charts load, **then** that tag is no longer counted. AC2. **Given** a symptom the therapist adds from the catalog, **when** the note and the charts load, **then** the symptom appears in both. AC3. **Given** a tag, **when** the therapist changes its intensity to another integer from 1 to 10, **then** the charts show the new value on their next load.

4.5.2 Evolution charts

Trend lines and heat maps of symptom intensity and frequency over time. CONOPS §4.5.2, CR-09.

US-22 See a symptom trend chart	AB#54
Statement	As a Therapist, I want to see a line chart of each symptom’s intensity across sessions, so that I can judge whether the patient is improving.
Priority	Should

US-22 (continued)

Platform	Web
Source	Client requirement; CR-09 (longitudinal charts: trend lines); CONOPS §4.5.2(b)
Depends on	US-20
Acceptance criteria	AC1. **Given** tagged notes on at least 2 dates, **when** the therapist opens the chart, **then** it shows a date axis and one line per symptom plotting its intensity. AC2. **Given** fewer than 2 tagged notes, **when** the therapist opens the chart, **then** "Not enough data" is shown. AC3. **Given** the date-range filter, **when** the chart first opens, **then** it covers the last 3 months (OQ-13) and the therapist can change the range. AC4. **Given** a 360 px wide screen, **when** the chart is shown, **then** it is fully readable without horizontal scrolling (CR-12).
Pending	Client confirmation of OQ-13

US-30 See a symptom heat map AB#78

Statement	As a Therapist, I want a heat map of symptom intensity per session, so that I can see which symptoms dominate and when.
Priority	Could
Platform	Web
Source	Client requirement; CR-09 (longitudinal heat maps); CONOPS §4.5.2(a)
Depends on	US-20
Acceptance criteria	AC1. **Given** tagged notes on at least 2 dates, **when** the therapist opens the heat map, **then** rows are symptoms, columns are session dates, cell color encodes intensity 1–10 with a legend, and a symptom not mentioned on a date shows an empty cell. AC2. **Given** the date-range filter, **when** the heat map first opens, **then** it covers the last 3 months (OQ-13). AC3. **Given** a cell, **when** the therapist hovers or taps it, **then** the symptom, date and intensity are shown. AC4. **Given** fewer than 2 tagged notes, **when** the therapist opens the heat map, **then** "Not enough data" is shown.
Pending	Client confirmation of OQ-13

4.6 Bibliographic cross-reference (future)

4.7 Deferred requirements

These stories are agreed with the client but are not part of this release (MoSCoW *Won't*).

US-23	**Get intervention suggestions from therapy manuals**	AB#55
Statement	As a Therapist, I want to cross-reference the patient's symptoms with therapy manuals, so that I receive suggestions for an intervention plan.	
Priority	Won't	
Platform	Web	
Source	Client requirement; CR-10 (bibliographic cross-reference, marked as future by the client)	
Depends on	—	
Acceptance criteria	AC1. No criteria yet: they are written only if the client includes this story in the project scope (OQ-19).	

4.8 Traceability

Table 4.1 links every story to its feature and to the client requirements it comes from. The backlog holds 30 stories: 13 *Must*, 13 *Should*, 3 *Could*, 1 *Won't*.

Table 4.1 Traceability of functional requirements

Story	Feature	Client req.	Priority	Platform	Depends on
US-01	F1.1	CR-02	Must	Web and Mobile	—
US-02	F1.1	CR-02	Must	Web and Mobile	US-01
US-24	F1.1	CR-02	Should	Web and Mobile	US-01
US-03	F1.2	CR-01	Must	Web	US-01
US-04	F1.2	CR-01	Should	Web	US-03
US-25	F1.2	CR-01	Should	Web	US-03
US-26	F1.2	—	Should	Web	US-03
US-27	F1.3	CR-11	Must	Web	US-01
US-28	F1.3	—	Should	Web	US-05
US-05	F2.1	CR-03	Must	Web	US-01
US-06	F2.1	CR-03	Must	Web and Mobile	US-05

Story	Feature	Client req.	Priority	Platform	Depends on
US-07	F2.1	CR-03	Must	Web	US-05
US-08	F2.1	CR-03	Must	Web	US-05
US-09	F3.1	CR-04	Should	Web	US-05
US-29	F3.1	—	Should	Mobile	US-09
US-10	F3.2	CR-04	Should	Web and Mobile	US-09
US-11	F3.3	CR-04	Should	Web and Mobile	US-10
US-12	F4.1	CR-05	Must	Web and Mobile	US-05
US-13	F4.1	CR-05	Should	Mobile	SPIKE-01, US-12, US-28
US-14	F4.1	CR-05	Could	Mobile	SPIKE-02, US-12, US-28
US-15	F4.2	CR-06	Must	Web and Mobile	US-12, US-28
US-16	F4.2	CR-06	Should	Web and Mobile	US-15
US-17	F4.2	CR-12	Must	Web and Mobile	US-15
US-18	F4.3	CR-07	Must	Web and Mobile	US-17
US-19	F4.4	—	Must	Web	US-12
US-20	F5.1	CR-08	Should	Web	US-17, US-28
US-21	F5.1	—	Could	Web	US-20
US-22	F5.2	CR-09	Should	Web	US-20
US-30	F5.2	CR-09	Could	Web	US-20
US-23	F6.1	CR-10	Won’t	Web	—

CHAPTER 5

NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements are classified according to Sommerville [5, §4.1.2] into product, organizational and external requirements. Each one has a *validation criterion*: a measurable test that tells whether the delivered system satisfies it [6]. Every criterion uses one of five methods: approval by an expert, a measurable test (over a period of time, with a number of users or with a number of test cases), a user acceptance test, a simulation, or a user satisfaction survey. Table 5.1 lists all of them; the cards below give the details.

Identifiers are `NFR-nn` and never change once published. A value followed by an open question (`OQ-nn`) is a team proposal that the client still has to confirm.

Table 5.1 Summary of non-functional requirements

ID	Requirement	Type	Validation
NFR-01	Post-session documentation time	Product, usability	User acceptance test
NFR-02	Responsive layout	Product, usability	Measurable test, cases
NFR-03	Interface language	Product, usability	Expert approval
NFR-04	Learnability	Product, usability	Test with users, survey
NFR-05	AI structuring time	Product, efficiency	Measurable test, cases
NFR-06	Patient search time	Product, efficiency	Measurable test, cases
NFR-07	Voice transcription time	Product, efficiency	Measurable test, cases
NFR-08	Screen response under load	Product, efficiency	Measurable test, users
NFR-09	Photo size limits	Product, efficiency (space)	Measurable test, cases

ID	Requirement	Type	Validation
NFR-10	Availability	Product, dependability	Measurable test, period
NFR-11	No loss of typed text	Product, dependability	Simulation
NFR-12	Backup and recovery	Product, dependability	Simulation
NFR-13	Non-enumerable identifiers	Product, security	Expert approval and cases
NFR-14	Encrypted connections	Product, security	Measurable test, cases
NFR-15	No patient identifiers sent to AI	Product, security	Expert approval and cases
NFR-16	Role separation enforced by the API	Product, security	Measurable test, cases
NFR-17	Password and session policy	Product, security	Measurable test, cases
NFR-18	Audit-log retention	Product, security	Expert approval
NFR-19	Technology stack and repository	Organizational, development	Expert approval
NFR-20	Deployment environment	Organizational, operational	Expert approval
NFR-21	Mobile platforms	Organizational, environmental	Measurable test, devices
NFR-22	Development process	Organizational, development	Expert approval
NFR-23	Personal-data protection law	External, legislative	Expert approval and cases
NFR-24	No automatic clinical decisions	External, ethical	Measurable test and expert
NFR-25	AI provider data terms	External, legislative	Expert approval

5.1 Product requirements

5.1.1 Usability

NFR-01	**Post-session documentation time**
Statement	A therapist shall complete the documentation of one session, from opening a new note to approving the structured note, in no more than 3 minutes.
Rationale	The client's main expected benefit is to return time to clinical work.

NFR-01 (continued)

Source	CR-12
Validation	Usability test with at least 5 participants, each documenting 3 sessions from synthetic session summaries of 150–250 words using the typed-text path; at least 90 % of the sessions are completed in ≤ 3 minutes, measured from tapping “New note” to “Approve”.

NFR-02	**Responsive layout**
Statement	Every screen of the web application shall be usable at widths of 360, 768 and 1280 pixels without horizontal scrolling or cut text.
Rationale	The client asks for a design that is fully responsive on computers, tablets and phones.
Source	CR-12
Validation	Measurable test with 30 test cases: the 10 web screens checked at the 3 widths in the latest versions of Chrome, Edge and Safari; all 30 cases pass.

NFR-03	**Interface language**
Statement	All texts, messages, dates and numbers of both applications shall be in Spanish as used in Ecuador (es-EC): dates as YYYY-MM-DD and decimal commas.
Rationale	The users are therapists in Guayaquil.
Source	Client’s document, written in Spanish; team proposal
Validation	Expert approval: the client reviews every screen of the prototype and of each sprint increment and finds no text in another language.

NFR-04	**Learnability**
Statement	After a demonstration of no more than 15 minutes (OQ-25), a therapist who has never used the system shall create a patient record and approve a note without help.
Rationale	Therapists have basic technical skills; the client asks for a minimalist and intuitive interface.
Source	CR-12
Validation	Measurable test with 3 therapists who did not take part in the design: all 3 finish both tasks without asking for help, and a satisfaction survey gives an average of at least 4 out of 5 for ease of use.

5.1.2 Efficiency

NFR-05	AI structuring time
Statement	The structured SOAP or DAP version of a note of up to 5,000 characters shall be shown within 20 seconds. After 30 seconds without an answer, the attempt counts as failed (OQ-21).
Rationale	Structuring must fit inside the 3-minute target of NFR-01.
Source	CR-06, CR-12
Validation	Measurable test with 20 synthetic notes between 500 and 5,000 characters: at least 18 are structured in ≤ 20 seconds and none takes more than 30 seconds without the failure message.

NFR-06	Patient search time
Statement	A search by name shall show its results within 1 second for a therapist with 500 patients (OQ-22).
Rationale	Finding the record is the first step of every session note.
Source	CR-03; team proposal
Validation	Measurable test with 50 searches on a database loaded with 500 synthetic patients: at least 95 % answer in ≤ 1 second.

NFR-07	Voice transcription time
Statement	A dictation of up to 5 minutes shall be returned as editable text within 60 seconds; dictations may last up to 10 minutes (OQ-08).
Rationale	Dictation is the capture path meant to save the most time after a session.
Source	CR-05
Validation	Measurable test with 10 synthetic recordings of 5 minutes in Spanish from Ecuador: at least 9 are transcribed in ≤ 60 seconds. The service is chosen in SPIKE-01.

NFR-08	Screen response under load
Statement	With 20 users working at the same time (OQ-28), the patient list, the patient record and the note history shall load in 2 seconds or less.
Rationale	The service is shared by every therapist of the clinic.
Source	Team proposal

Validation	Measurable test with 500 requests generated by a load-testing tool simulating 20 concurrent users: the 95th percentile of the response time is ≤ 2 seconds.

NFR-09 Photo size limits	
Statement	A note shall accept up to 5 photos in JPG or PNG format of up to 10 MB each (OQ-09); larger files are rejected with a message that states the limit.
Rationale	Limits the storage and processing cost of handwritten notes.
Source	`CR-05`
Validation	Measurable test with 8 test cases: 5 photos of 10 MB are accepted; a sixth photo, an 11 MB file and a PDF file are each rejected with the message.

5.1.3 Dependability

NFR-10 Availability	
Statement	The system shall be available at least 99 % of the time between 07:00 and 21:00, Monday to Saturday, Ecuador time (OQ-26).
Rationale	The client only states that the system is hosted on a VPS, which is not measurable. The proposed window covers the clinic's working hours.
Source	`CR-13`
Validation	Measurable test over a period of 30 days: an external monitor checks the sign-in page every 5 minutes, and the share of successful checks inside the window is at least 99 %.

NFR-11 No loss of typed text	
Statement	When the AI service, the transcription service or the network fails, the text already typed or transcribed shall be kept as a draft, and the therapist shall be told what failed.
Rationale	A failure of an external service must not cost the therapist the note.
Source	`CR-06`; team proposal
Validation	Simulation with 10 test cases: the AI service is made unavailable or slower than 30 seconds while a note is being structured; in all 10 cases the draft is intact after reloading the page.

NFR-12	Backup and recovery
Statement	The database shall be backed up every day. After a server failure, at most the data of the last 24 hours is lost and the service is restored within 4 hours (OQ-27).
Rationale	Clinical records cannot be rebuilt if they are lost.
Source	Team proposal
Validation	Simulation: the previous day's backup is restored on a clean server; the client confirms that the synthetic records are complete, and the time from start to a working sign-in page is ≤ 4 hours.

5.1.4 Security

NFR-13	Non-enumerable identifiers
Statement	Users, patients, appointments and notes shall be identified by UUIDs, never by sequential numbers, in the database, the API and every address.
Rationale	Prevents guessing the address of another patient's record.
Source	`CR-11`
Validation	Expert approval of the database schema by the client, and a measurable test with 20 cases: changing the identifier in an address or an API call never returns another user's record.

NFR-14	Encrypted connections
Statement	Every connection between the applications, the API and external services shall use HTTPS with TLS 1.2 or later; plain HTTP requests are redirected to HTTPS.
Rationale	Health data travels over the internet.
Source	`CR-11`
Validation	Measurable test with 50 requests: 100 % of HTTP requests are redirected, and an SSL test of the domain reports a valid certificate and no protocol older than TLS 1.2.

NFR-15	No patient identifiers sent to the AI service
Statement	Names, identity numbers, phone numbers and emergency contacts of patients shall not be sent to the external AI, transcription or OCR services (OQ-17).
Rationale	Limits the personal data that leaves the clinic.

NFR-15 (continued)

Source	`CR-11`; [8]
Validation	Expert approval and measurable test with 20 synthetic notes that mention the patient's name: a reviewer inspects the log of outgoing requests and finds no identifier in any of them.

NFR-16 Role separation enforced by the API

Statement	The API shall reject with 403 every request for clinical content (notes, symptoms, charts) made by a Clinic Administrator, and every request by a therapist for another therapist's patients (OQ-04, OQ-18).
Rationale	Hiding an option in the interface is not enough: permissions must hold for any client.
Source	`CR-01`
Validation	Measurable test with 20 API calls made directly with the token of each role: all forbidden calls answer 403 and return no data.

NFR-17 Password and session policy

Statement	Passwords shall have at least 10 characters (OQ-23) and be stored only as salted hashes; an account is locked for 15 minutes after 5 consecutive failed sign-ins (OQ-01); a session expires after 30 minutes without activity (OQ-02).
Rationale	The sign-in is the only barrier in front of mental-health records.
Source	`CR-02`
Validation	Measurable test with 10 test cases covering each rule, plus expert approval of the user table: no password is readable in it.

NFR-18 Audit-log retention

Statement	Every access to a patient record shall be logged with user, date, time and action, and the log shall be kept for at least 5 years and be readable only by the Clinic Administrator (OQ-16).
Rationale	Allows the clinic to answer who saw a record.
Source	US-27; team proposal
Validation	Expert approval: the client reviews the retention setting and the log produced by 10 test accesses, and confirms that no role can edit or delete an entry.

5.2 Organizational requirements

NFR-19	Technology stack and repository
Statement	The web application shall be built with React, TypeScript and Tailwind CSS; the API with FastAPI (Python); the data shall be stored in PostgreSQL through SQLAlchemy; and the clients and the API shall live in separate folders of one repository.
Rationale	The client fixed this technology so the system can be maintained by their own staff.
Source	`CR-14` to `CR-17`
Validation	Expert approval: the client reviews the repository at the end of each sprint.

NFR-20	Deployment environment
Statement	The system shall run on the client's Ubuntu VPS, with Nginx as web server and reverse proxy with SSL, the API as a systemd service, and PostgreSQL on port 5433.
Rationale	It is the infrastructure the client already operates.
Source	`CR-17`, `CR-18`
Validation	Expert approval: the client verifies the installation on the VPS at the end of the project.

NFR-21	Mobile platforms
Statement	The mobile application shall run on Android 10 or later and iOS 15 or later (OQ-29), on phones and tablets.
Rationale	Covers the devices therapists are expected to carry.
Source	CONOPS §4.2; team proposal
Validation	Measurable test on 4 devices (one phone and one tablet per platform, at the minimum versions): the four mobile screens of Appendix B work on all of them.

NFR-22	Development process
Statement	The team shall manage the work with Scrum in Azure DevOps: every change reaches the main branch through a pull request reviewed by another member and linked to a work item.
Rationale	Keeps the requirements, the code and their evidence linked for the client and the course.

NFR-22 (continued)

Source	Course rules; team working agreement
Validation	Expert approval: the instructor and the client review the repository history at each sprint review; no commit reaches the main branch without a pull request.

5.3 External requirements

NFR-23	**Personal-data protection law**
Statement	The system shall record each patient's informed consent before any of their notes is processed by an external AI service, and shall stop that processing when the consent is withdrawn (OQ-15).
Rationale	Ecuador's Organic Law on Personal Data Protection treats health data as a special category that needs explicit consent [8].
Source	US-28; [8]
Validation	Expert approval by the client's legal adviser, and a measurable test with 20 cases: no note of a patient without consent can be dictated, photographed, structured or tagged.

NFR-24	**No automatic clinical decisions**
Statement	The system shall not issue diagnoses or treatment recommendations. Everything the AI produces stays as a draft until the therapist approves it.
Rationale	The professional is responsible for the clinical record.
Source	CONOPS §2.2.2; team working agreement
Validation	Measurable test with 20 test cases and expert approval: the client confirms that no note reaches the "Approved" state without the therapist's action and that no screen shows a diagnosis.

NFR-25	**AI provider data terms**
Statement	The AI service shall be used under terms that do not allow the provider to train its models on the data sent (OQ-17).
Rationale	Clinical text must not become part of a third party's model.
Source	`CR-16`; [8]
Validation	Expert approval: the client reviews the provider's contract or data-processing terms before the system goes into operation.

CHAPTER 6

EXTERNAL INTERFACE REQUIREMENTS

6.1 User interfaces

Both applications use one visual system: a warm off-white background, slate text, a single dark teal accent reserved for actions and the current selection, line icons instead of text where an icon is clear, and labels in sentence case. All texts are in Spanish (es-EC, NFR-03). Table 6.1 lists the screens of the prototype; their images and the window flow are in Appendix B.

Table 6.1 Screens of the prototype

Screen	Purpose	User stories
W01	Sign in, with Google and Microsoft accounts shown for a future version	US-01, US-02, US-24
W02	Patient list with search, active or archived filter and dropout tag	US-06, US-08, US-11
W03	Create or edit a patient record, including AI consent	US-05, US-07, US-28
W04	Patient record: data, attendance figures, edit and archive	US-07, US-08, US-10, US-11
W05	Appointments of a patient and their attendance	US-09, US-10
W06	New note by text, voice or photo, with the previous session's summary	US-12 to US-16, US-18
W07	Review the structured note next to the original text and its symptom tags	US-17, US-20, US-21
W08	Note history, without a delete option, with addenda	US-19
W09	Symptom evolution chart with a time range	US-21, US-22
W10	Staff accounts and roles (Clinic Administrator)	US-03, US-04, US-25
M01	Today's agenda on the phone	US-29, US-10, US-11
M02	Patient summary and the previous session	US-06, US-18
M03	New note by voice dictation, text or photo	US-12, US-13, US-14
M04	Review and approve the structured note	US-15, US-16, US-17

Source: Prepared by the team

6.2 Software interfaces

Table 6.2 Software interfaces

Interface	Purpose	Data exchanged	Failure handling
OpenAI API (language model)	Structure notes into SOAP or DAP; extract and tag symptoms; summarize the previous session	Sends the note text without patient identifiers (NFR-15); receives the structured sections and the tags as JSON	After 30 seconds without an answer the attempt fails; the draft and the original text are kept (NFR-05, NFR-11)
Speech-to-text service	Transcribe dictated notes (US-13)	Sends audio of up to 10 minutes; receives text. The audio is deleted after transcription (OQ-08)	"No speech detected" or a retry message; typed text is unchanged
Handwriting OCR service	Read photos of handwritten notes (US-14)	Sends up to 5 images; receives text	"No readable text" message; the photos are not stored
PostgreSQL	Store users, patients, appointments, notes, tags and the audit log	SQL through SQLAlchemy, port 5433, only from the API	Daily backups (NFR-12)

Source: Prepared by the team from `CR-06`, `CR-16` and `CR-17`

The speech-to-text and OCR services are chosen in SPIKE-01 and SPIKE-02 by comparing at least two options on accuracy, response time and cost.

6.3 Hardware interfaces

The system has no special hardware. The mobile application uses the device's microphone for dictation and its camera for photos of handwritten notes, only after the user grants permission. The web application can use the computer's microphone in the same way.

6.4 Communication interfaces

- Every client talks to the API over HTTPS with TLS 1.2 or later (NFR-14); Nginx redirects plain HTTP to HTTPS.

- The API exposes a REST interface that exchanges JSON. Both clients call it through a typed `httpClient.ts` wrapper (`CR-15`).
- After sign-in, each request carries a JWT bearer token (`CR-02`). The API answers 401 when the token is missing, expired or revoked, and 403 when the role is not allowed (NFR-16).
- The API is the only component that talks to the database and to the external services.

CHAPTER 7

GLOSSARY

Table 7.1 Glossary of terms and acronyms

Term	Meaning
Addendum	Dated text added to an approved note to correct or complete it without changing the original content.
Approved note	Clinical note that the therapist reviewed and approved. Only approved notes feed the previous-session summary, the symptom tags and the charts.
Concept of operations	Document that describes the stakeholders, the current situation and the proposed system from the users' point of view.
Context memory	Summary of the previous session that the system shows when a new note is started for the same patient.
DAP note	Clinical note format with three sections: Data, Assessment and Plan.
Digital patient record	Electronic record holding a patient's demographic data, history, appointments and clinical notes.
Dropout alert	Warning shown when a patient is at high risk of abandoning treatment.
Draft note	Note that has not been approved yet. AI output is always a draft.
Heat map	Chart that uses color to show the intensity or frequency of a patient's symptoms over time.

Term	Meaning
Open question	Value proposed by the team that the client still has to confirm, identified as `OQ-nn`.
Product backlog	Ordered list of everything the product needs, kept in Azure Boards as epics, features and user stories.
Responsive design	Design approach in which the interface adapts to computers, tablets and phones.
SOAP note	Clinical note format with four sections: Subjective, Objective, Assessment and Plan.
Spike	Short, time-boxed investigation that reduces the technical uncertainty of a story before it is committed to a sprint.
User authentication	Process that verifies a user's identity to control access to functions or roles.
Validation criterion	Measurable test that tells whether the delivered system satisfies a requirement (fit criterion).
AI	Artificial intelligence.
API	Application programming interface.
ESPOL	Escuela Superior Politécnica del Litoral.
HTTPS	Hypertext Transfer Protocol Secure.
JWT	JSON Web Token.
LOPDP	Ley Orgánica de Protección de Datos Personales (Ecuador's Organic Law on Personal Data Protection).
MoSCoW	Prioritization scheme: Must, Should, Could, Won't.
NFR	Non-functional requirement.
OCR	Optical character recognition.
RBAC	Role-based access control.
REST	Representational state transfer: style of web API that exchanges resources over HTTP.

Term	Meaning
SaaS	Software as a service.
TLS	Transport Layer Security.
UUID	Universally unique identifier.
VPS	Virtual private server.

Source: Adapted from CogniSync Team 1 [1, Appendix A.2]

REFERENCES

[1] CogniSync Team 1, *Cognisync — project specification (concept of operations), version 1.0*, ESPOL, Software Engineering I, Oct. 2026.

[2] *Systems and software engineering — life cycle processes — requirements engineering*, ISO/IEC/IEEE, 2018. DOI: 10.1109/IEEESTD.2018.8559686

[3] *IEEE recommended practice for software requirements specifications*, IEEE, 1998. DOI: 10.1109/IEEESTD.1998.88286

[4] M. Cohn, *User Stories Applied: For Agile Software Development*. Addison-Wesley, 2004.

[5] I. Sommerville, *Software Engineering*, 10th ed. Pearson, 2016.

[6] S. Robertson and J. Robertson, *Mastering the Requirements Process: Getting Requirements Right*, 3rd ed. Addison-Wesley, 2012.

[7] Atlantic Systems Guild. "Volere requirements specification template," Accessed: Oct. 26, 2026. [Online]. Available: https://www.volere.org/templates/volere-requirements-specification-template/

[8] Asamblea Nacional del Ecuador, *Ley orgánica de protección de datos personales*, Registro Oficial Suplemento No. 459, May 2021.

[9] CogniSync Team 1. "Cognisync public repository." Elicitation evidence and prototype, Accessed: Oct. 26, 2026. [Online]. Available: https://github.com/mateoandraded/cognisync

[10] Mobbin. "Mobbin: UI and UX design reference library." Screens of Heidi, Motion, Sora, Lyssna and Hotjar consulted as layout references, Accessed: Nov. 3, 2026. [Online]. Available: https://mobbin.com

INDIVIDUAL CONTRIBUTIONS

Each row lists the work that the repository history, Azure Boards and the elicitation log attribute to each member.

TODO: Each member confirms or completes their own row before submission. Andrés Saltos's row has no recorded evidence yet.

Member	Contribution
Mateo Andrés Andrade Domínguez	Azure DevOps organization and project; first product backlog (23 stories in 6 epics, 2 spikes, items #27 to #57) and the questionnaire for the client; this version of the SRS: context diagram, feature model, Chapters 3, 5 and 6, Appendix B; navigable prototype of the web and mobile applications.
Juan Francisco Fernández Ramos	Repository foundation and working agreement; English transcription of the client requirements and the open questions; backlog v2 (Given/When/Then criteria, 14 features, stories US-24 to US-30); SRS skeleton, LaTeX class and the Boards-to-LaTeX export.
Victor Manuel Morales Vásquez	Led the refinement meeting with the client on 2026-10-30, reading each user story with him.
Diego Xavier Parrales Villacreses	Concept of operations template and first repository content (2026-10-16).
Andrés Alexander Saltos Preciado	

AUTHORSHIP DECLARATION

The members of Team 1 declare that this document is their own work and an accurate representation of their intellectual contribution, that it contains no plagiarized or falsified content, and that they take responsibility for the accuracy of all of it, including the parts produced with generative artificial intelligence tools. This declaration follows the course's authorship policy, based on the ACM policy on authorship.

Table 7.2 discloses every substantial use of generative AI, as recorded in the team's AI usage log (`docs/process/ai-usage-log.md`). Uses limited to spelling and grammar are not listed.

Table 7.2 Use of generative AI tools

Member, date	Tool	Parts produced	Prompt summary
Mateo Andrade, 2026-10-26	Claude Code (Claude Opus 5.5)	First draft of the 23 user stories, their acceptance criteria and the questionnaire for the client (Spanish)	Write user stories with acceptance criteria and MoSCoW priority from the client's document
Juan Francisco Fernández, 2026-10-26	Claude Code (Claude Opus 5.5)	Repository foundation, English transcription of the client requirements, open questions, product vision	Organize the project around Azure DevOps and prepare it for the midterm SRS
Juan Francisco Fernández, 2026-10-26	Claude Code (Claude Opus 5.5)	Backlog v2: English translation, Given/When/Then criteria, 14 features, 7 gap stories	Apply the instructor's feedback while keeping the team's proposed values
Juan Francisco Fernández, 2026-10-26	Claude Code (Claude Opus 5.5)	SRS skeleton: LaTeX class, draft of Chapters 1 and 2, Section 3 methodology, NFR-01, glossary, export script	Prepare the midterm SRS with functional requirements generated from Azure Boards

Member, date	Tool	Parts produced	Prompt summary
Mateo Andrade, 2026-11-03	Claude Code (Claude Opus 5.5), Hallmark design skill, Mobbin	Context diagram, feature model, Chapter 3, NFR-02 to NFR-25, Chapter 6, Appendix B, drafts of these two closing sections, navigable prototype	Port the team's Spanish SRS draft into the LaTeX skeleton and improve it; redesign the prototype with a restrained palette and icons, using Mobbin screens as references

Human review. The team reviewed every AI-produced part in pull requests before merging it: proposed values were turned into open questions for the client instead of being presented as confirmed, the user stories were checked with the client in the refinement meeting of 2026-10-30, and the prototype was compared against the acceptance criteria.

TODO: Each member reads this declaration and states their agreement in the pull request. Add any other AI use that is not in the log.

APPENDIX A

CLIENT CONFORMITY ACT

TODO: Place the signed conformity act at `docs/srs/private/conformity-act.pdf` (never committed). Without it the deliverable loses 100 points.

APPENDIX B

PROTOTYPE

The prototype is a navigable, non-functional model of both applications, written in HTML and CSS without business logic. It has 10 web screens and 4 mobile screens that cover 26 of the 29 stories of this release; US-26 (staff and patient lists), US-27 (audit log) and US-30 (heat map) have no screen yet. It lets the client check the content of each screen and the flow between them before any code is written. It uses synthetic data only: no name, date or note belongs to a real patient. Its files are in the public repository [9], and it can be tried online at `https://mateoandraded.github.io/cognisync/`. Table 6.1 in Chapter 6 maps each screen to its user stories.

B.1 Window flow

Figure B.1 shows how the screens connect. A therapist signs in and reaches the patient list; from a patient's record they open the appointments, a new note, the note history or the evolution chart. A Clinic Administrator signs in and reaches only the staff screen, without access to clinical content (NFR-16). On the phone, the therapist goes from today's agenda to the patient, the dictation and the review of the note, and returns to the agenda after approving it.

B.2 Web application screens

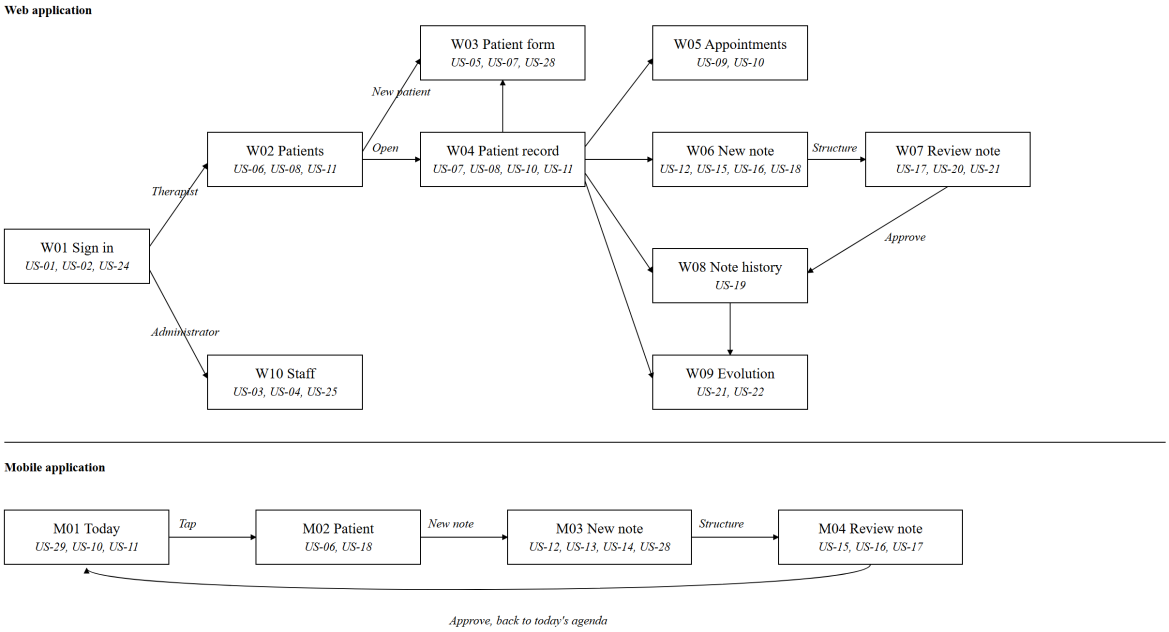


Figure B.1 Window flow of the web and mobile applications
Source: Prepared by the team

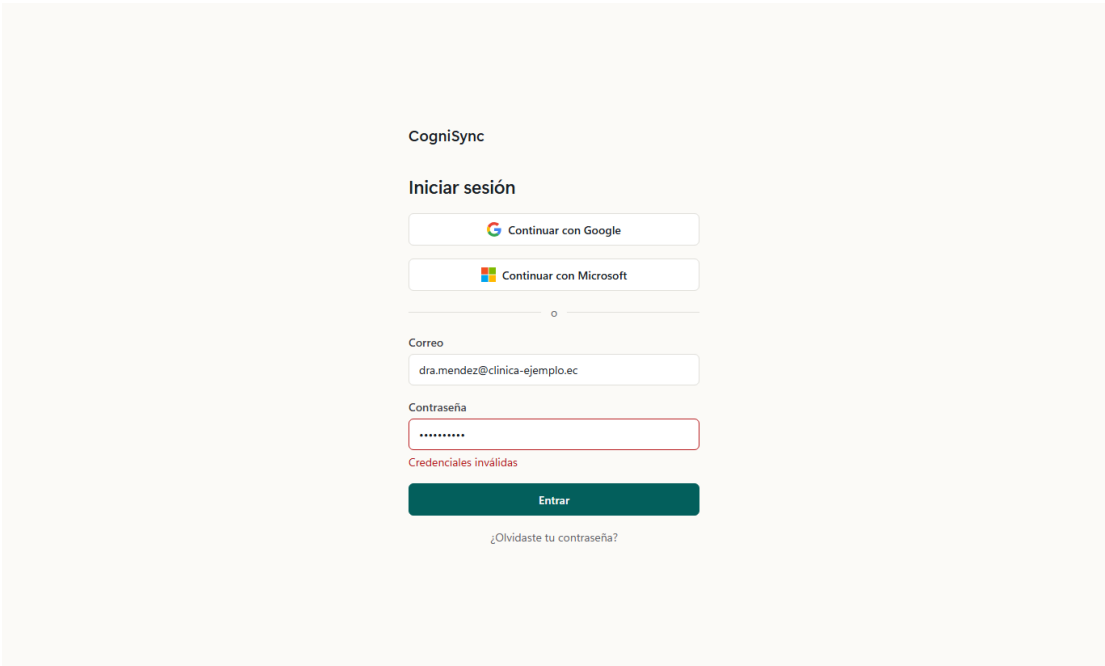


Figure B.2 Web, W01: sign in, showing the invalid-credentials message (US-01)
Source: Prototype, synthetic data

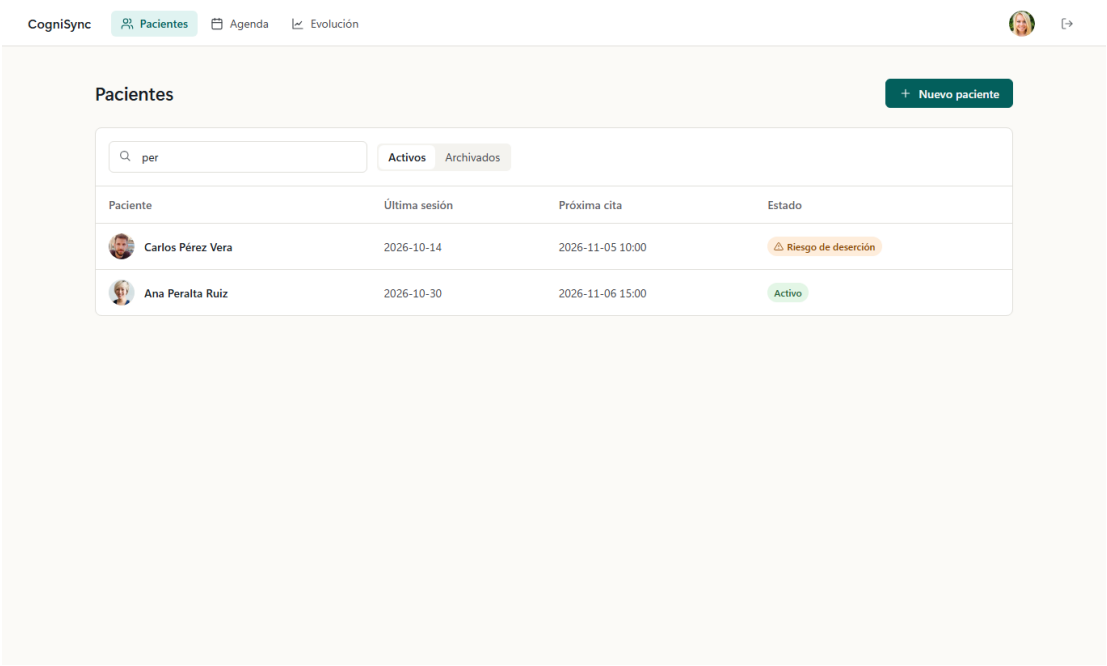


Figure B.3 Web, W02: patient list with search and dropout-risk tag (US-06, US-11)
Source: Prototype, synthetic data

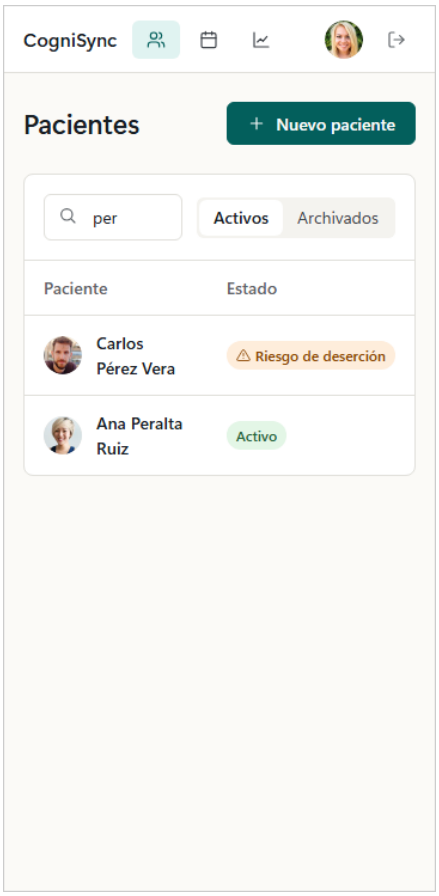


Figure B.4 Web, W02 at 360 pixels wide, without horizontal scrolling (NFR-02)
Source: Prototype, synthetic data

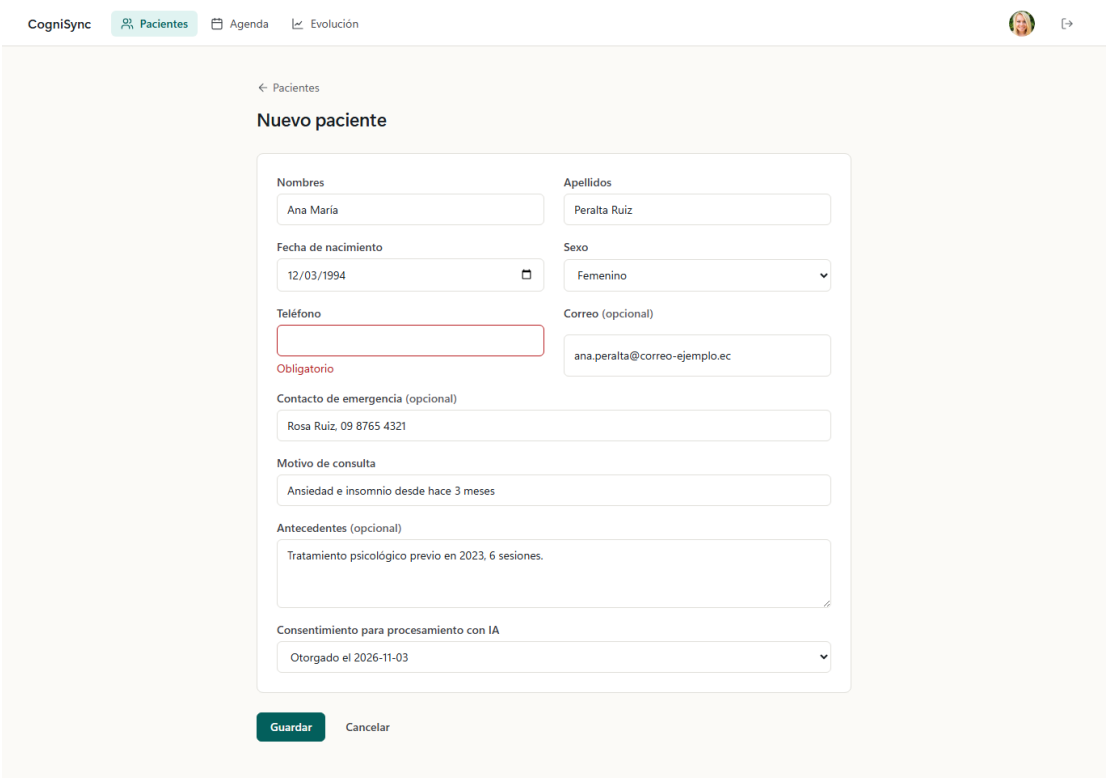


Figure B.5 Web, W03: patient form with a missing required field and the AI consent (US-05, US-28)
Source: Prototype, synthetic data

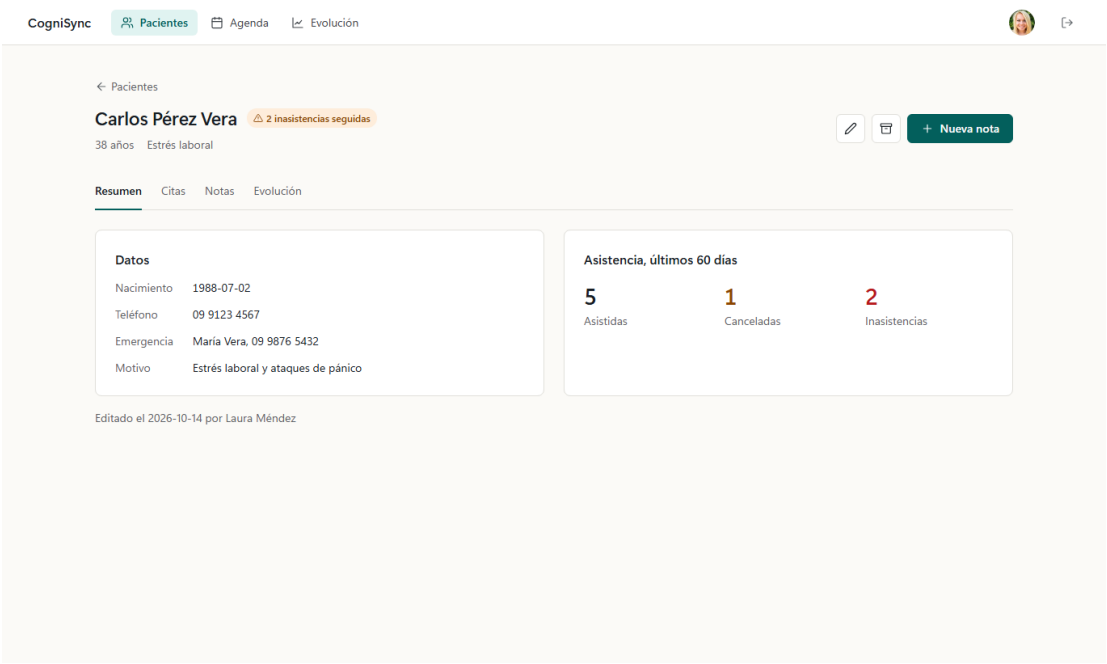


Figure B.6 Web, W04: patient record with attendance figures and dropout alert (US-07, US-10, US-11)
Source: Prototype, synthetic data

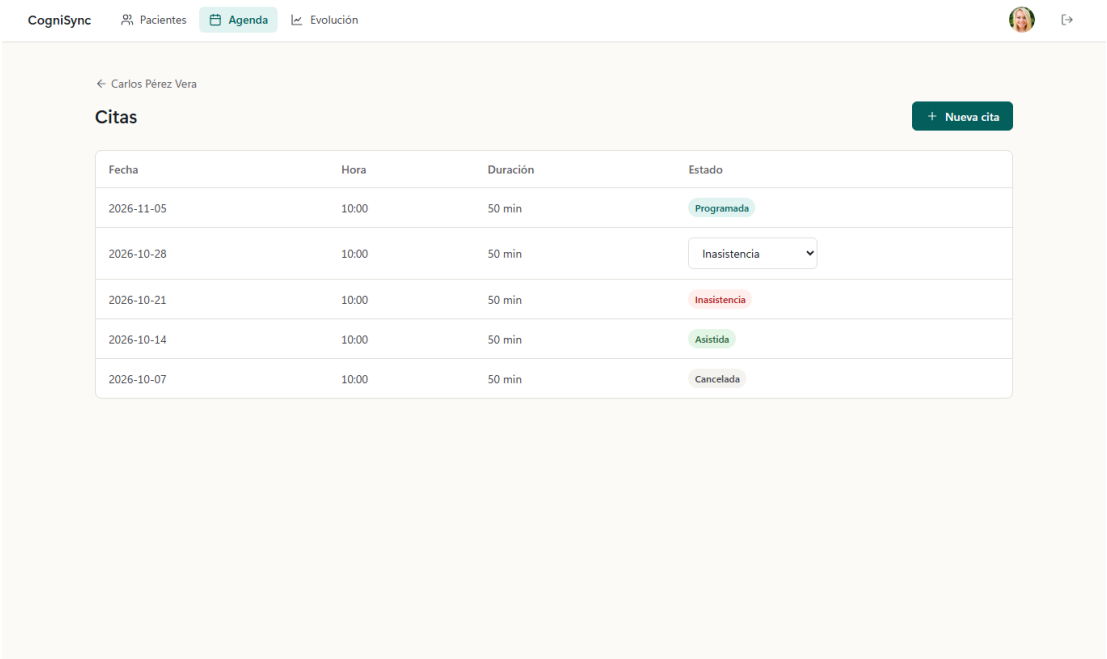


Figure B.7 Web, W05: appointments and attendance of a patient (US-09, US-10)
Source: Prototype, synthetic data

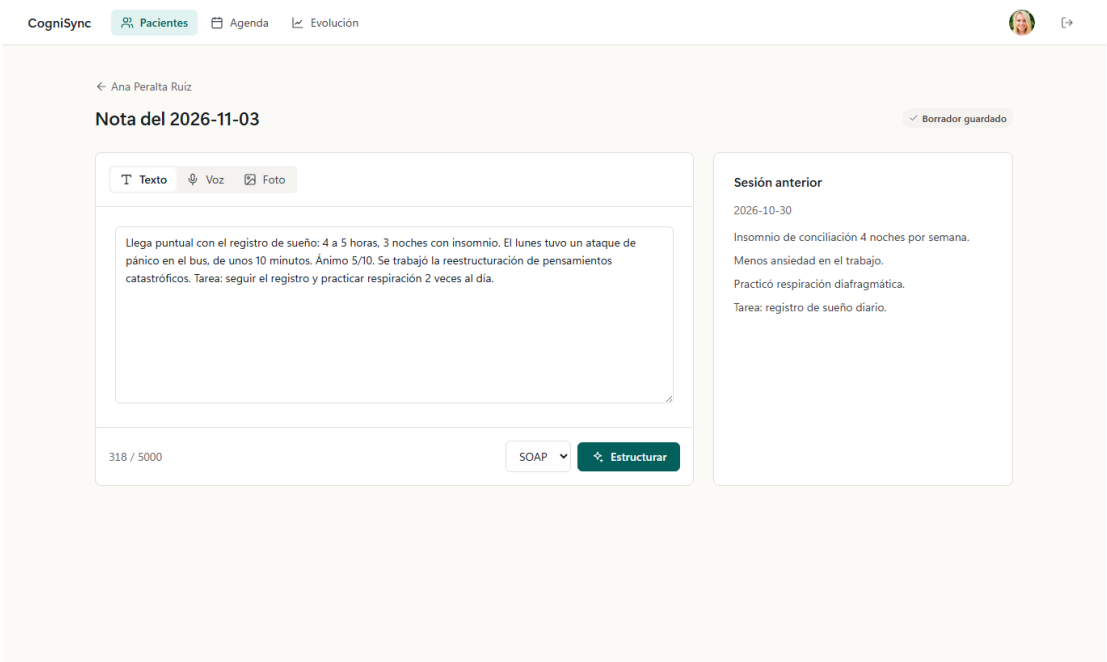


Figure B.8 Web, W06: new note with the previous session's summary (US-12, US-18)
Source: Prototype, synthetic data

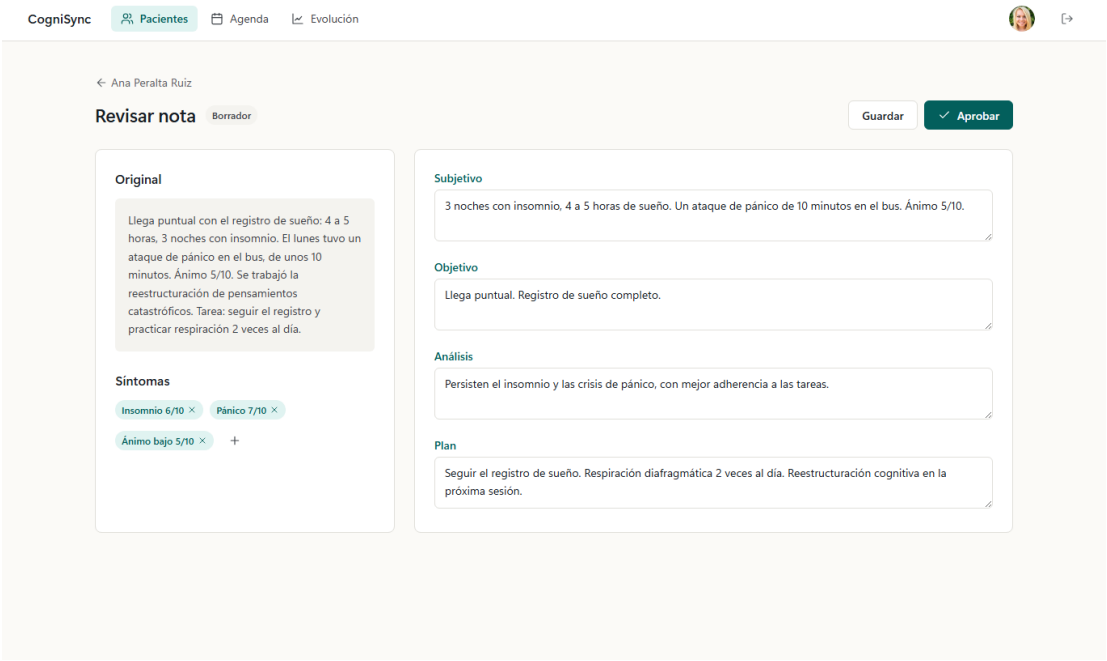


Figure B.9 Web, W07: structured SOAP note next to the original text, with symptom tags (US-17, US-20, US-21)
Source: Prototype, synthetic data

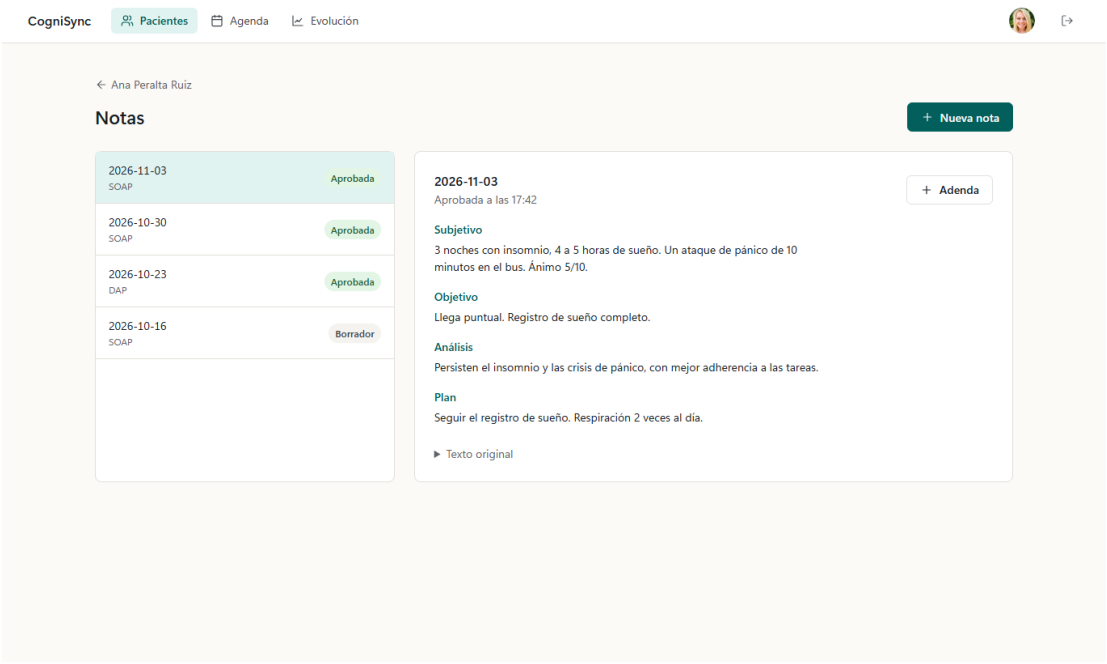


Figure B.10 Web, W08: note history without a delete option (US-19)
Source: Prototype, synthetic data

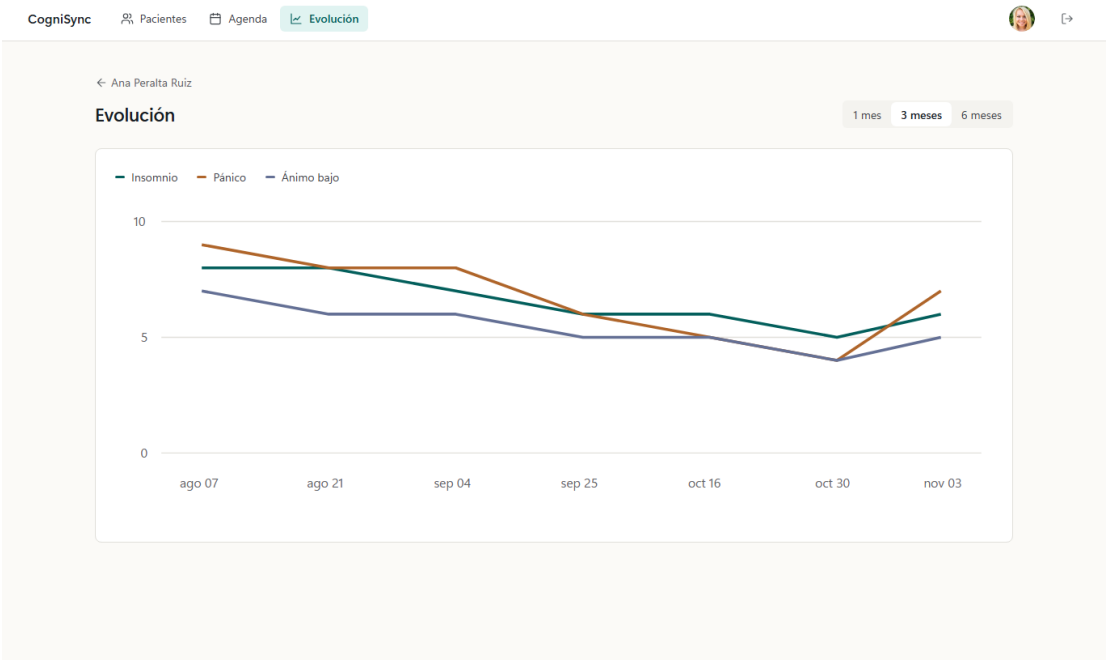


Figure B.11 Web, W09: symptom evolution chart over three months (US-22)
Source: Prototype, synthetic data

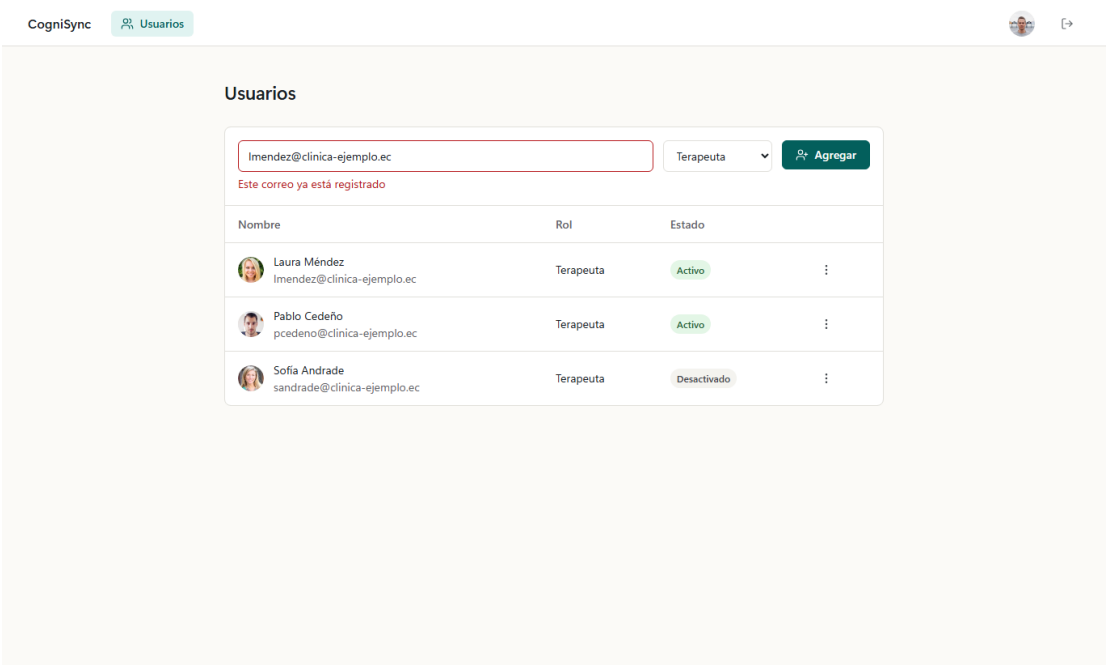
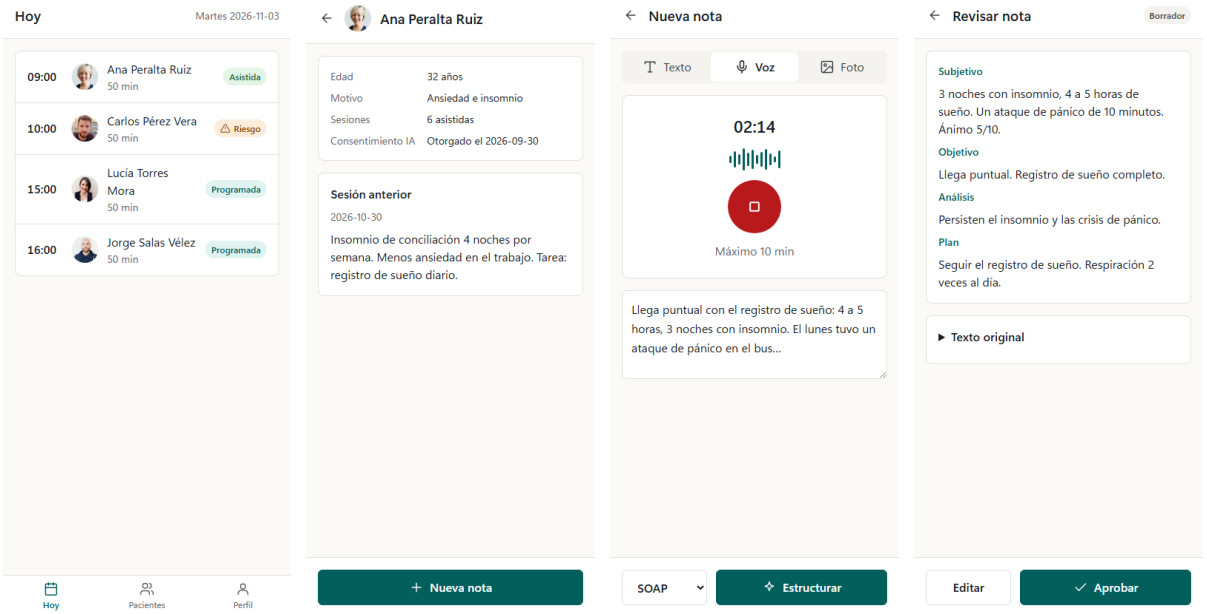


Figure B.12 Web, W10: staff accounts, rejecting a duplicate email (US-03, US-04)
Source: Prototype, synthetic data

B.3 Mobile application screens



(a) M01: today's agenda (US-29)

(b) M02: patient and previous session (US-18)

(c) M03: voice dictation (US-13)

(d) M04: review and approve (US-17)

Figure B.13 Mobile application screens
Source: Prototype, synthetic data